THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 11: **World of Algorithms** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Adding 1747 and 4333 by the column algorithm, the number of columns in which a carry is generated is: 1

(a) 2 **(b)** 3 **(c)** 1 **(d)** 4

2. The divisor algorithm applied to 44 produces a list with how many entries? 1

(a) 8 **(b)** 7 **(c)** 6 **(d)** 5

3. By Euclid's algorithm, gcd$(90, 82)$ is: 1

(a) 4 **(b)** 2 **(c)** 3 **(d)** 1

4. The value of $146 \bmod 3$ (remainder on dividing 146 by 3) is: 1

(a) 2 **(b)** 4 **(c)** 1 **(d)** 3

5. In the divisor algorithm for n, the numbers j that are checked are: 1

(a) $1, 2, \ldots, n$ **(b)** only the primes up to n

(c) $2, 3, \ldots, n-1$ only **(d)** only the even numbers up to n

6. Using Euclid's subtraction algorithm for gcd$(90, 50)$, the number of subtraction steps needed is: 1

(a) 5 **(b)** 6 **(c)** 8 **(d)** 7

7. Set x = 8. Repeat 6 times: x = 2x − 1. The final value of x is: 1

(a) 449 **(b)** 451 **(c)** 448 **(d)** 450

8. By Euclid's algorithm, gcd$(138, 117)$ is: 1

(a) 4 **(b)** 2 **(c)** 5 **(d)** 3

9. Adding 1497 and 3340 by the column algorithm, the number of columns in which a carry is generated is: 1

(a) 1 **(b)** 0 **(c)** 2 **(d)** 3

10. The divisor algorithm applied to 77 produces a list with how many entries? 1

(a) 4 **(b)** 3 **(c)** 5 **(d)** 6

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Apply Euclid's subtraction algorithm to find gcd(54, 30). Write the sequence of pairs and count the subtractions. 2

12. Apply Euclid's subtraction algorithm to find gcd(70, 42). Write the sequence of pairs and count the subtractions. 2

13. Run the divisor algorithm for 42: check j = 1, 2, …, 42. Write the list of divisors and say how many numbers are checked. 2

14. Apply Euclid's subtraction algorithm to find gcd(55, 40). Write the sequence of pairs and count the subtractions. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Apply Euclid's subtraction algorithm to find gcd(92, 72). Write the sequence of pairs and count the subtractions. 3

16. Apply Euclid's subtraction algorithm to find gcd(195, 165). Write the sequence of pairs and count the subtractions. 3

17. Add 2860 and 7672 using the column algorithm. State in which columns a carry is produced. 3

OR

Apply Euclid's subtraction algorithm to find gcd(112, 98). Write the sequence of pairs and count the subtractions. 3

18. Apply Euclid's subtraction algorithm to find gcd(165, 135). Write the sequence of pairs and count the subtractions. 3

OR

Add 5199 and 3314 using the column algorithm. State in which columns a carry is produced. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Trace: x = 5; repeat 5 times: x = 2x − 2. Write the value of x after each repetition and the final value. 5

OR

Trace: s = 0; while n > 0, add (n mod 10) to s and replace n by its quotient on division by 10. Start with n = 5696. Write the digits picked and the final s. 5

20. Trace: x = 9; repeat 4 times: x = 2x − 1. Write the value of x after each repetition and the final value. 5

OR

Trace: s = 0; while n > 0, add (n mod 10) to s and replace n by its quotient on division by 10. Start with n = 4860. Write the digits picked and the final s. 5

END OF QUESTION PAPER — Total: 40 Marks